

VLISSINGEN, Netherlands

WMO: 063100

Lat: 51.4414N

Lon: 3.5958E

Elev: 10

StdP: 101.20

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.8	-2.1	-7.5	2.0	-2.5	-5.5	2.4	-1.4	19.0	9.4	17.5	9.0	5.3	60	0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.6	26.7	19.9	24.7	19.0	23.0	18.2	20.6	25.0	19.7	23.5	18.9	22.1	4.2	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	13.9	22.7	18.2	13.1	21.7	17.5	12.6	20.9	59.4	25.2	56.4	23.5	53.7	22.2	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.6	13.7	12.1	-4.9	30.9	2.7	2.3	-6.9	32.5	-8.5	33.9	-10.0	35.2	-12.0	36.8		
(5)			-5.7	22.5	2.6	0.8	-7.6	23.0	-9.1	23.5	-10.6	24.0	-12.5	24.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	4.5	4.7	6.9	10.0	13.5	16.3	18.6	18.8	16.2	12.7	8.6	5.4		
	DBStd	5.88	3.42	3.13	2.75	3.01	3.02	2.62	2.68	2.34	2.24	2.68	2.75	3.38		
	HDD10.0	668	172	147	100	35	4	0	0	0	0	8	56	146		
	HDD18.3	2628	429	381	354	249	154	73	28	21	71	176	291	402		
	CDD10.0	1175	1	1	4	36	112	188	267	272	187	91	16	2		
	CDD18.3	93	0	0	0	0	3	11	37	34	8	1	0	0		
	CDH23.3	333	0	0	0	1	9	46	151	114	13	0	0	0		
	CDH26.7	61	0	0	0	0	1	7	33	20	1	0	0	0		
(14) Wind	WSAvg	5.9	7.1	6.6	6.2	5.4	5.4	5.3	5.2	5.2	5.4	6.2	6.4	6.9		
(15) Precipitation	PrecAvg	743	57	46	50	44	50	63	66	74	71	77	79	73		
	PrecMax	969	146	102	102	82	99	124	166	203	211	236	146	159		
	PrecMin	430	3	2	9	8	5	5	12	7	4	7	18	15		
	PrecStd	124	28	26	25	20	26	32	34	45	42	46	30	36		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	11.7	15.8	21.7	24.4	27.4	29.9	29.1	25.1	20.2	15.7	12.7		
		MCWB	10.2	9.3	11.5	14.2	16.8	19.8	20.7	20.4	19.6	16.8	14.0	11.3		
	2%	DB	10.5	10.5	13.1	18.7	22.2	24.3	26.9	26.3	22.5	18.2	14.2	11.6		
		MCWB	9.3	8.9	9.9	13.3	16.3	18.4	19.8	20.2	18.2	15.8	12.8	10.4		
	5%	DB	9.4	9.5	11.5	16.2	20.0	22.1	24.7	24.2	20.8	17.1	13.3	10.7		
		MCWB	8.3	8.1	9.2	11.9	15.0	17.6	19.1	19.1	17.4	15.2	11.9	9.6		
	10%	DB	8.6	8.7	10.3	14.2	18.0	20.2	22.7	22.3	19.5	16.2	12.3	9.7		
		MCWB	7.6	7.5	8.6	10.9	14.2	16.4	18.3	18.4	16.6	14.5	10.9	8.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	10.1	11.7	15.2	18.8	20.6	21.8	21.8	20.1	17.2	14.2	11.6		
		MCDB	11.3	11.3	14.5	20.2	23.5	25.5	27.1	26.5	23.7	19.3	15.1	12.3		
	2%	WB	9.5	9.1	10.5	13.5	16.7	19.1	20.6	20.7	18.9	16.3	13.1	10.6		
		MCDB	10.3	10.1	12.4	17.6	20.4	23.2	25.2	24.9	21.5	17.9	14.1	11.4		
	5%	WB	8.5	8.3	9.6	12.3	15.5	17.9	19.7	19.7	18.0	15.5	12.1	9.7		
		MCDB	9.2	9.2	11.0	15.4	18.9	21.3	23.8	23.0	20.2	16.9	13.2	10.5		
	10%	WB	7.7	7.6	8.9	11.4	14.5	16.9	18.8	18.9	17.2	14.7	11.2	8.7		
		MCDB	8.4	8.5	10.1	13.9	17.5	19.7	22.1	21.8	19.2	16.0	12.1	9.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.3	3.7	4.6	5.7	5.9	5.9	5.8	5.6	5.0	4.2	3.4	3.3		
		MCDBR	3.6	4.3	6.5	8.7	9.2	9.3	9.1	8.5	6.8	4.8	3.7	3.5		
	5% WB	MCWBR	3.5	3.4	4.3	4.6	4.7	4.9	4.1	4.0	3.7	3.4	3.3	3.4		
		MCWBR	3.6	3.9	5.5	7.8	8.2	8.1	8.0	7.6	6.0	4.6	3.6	3.5		
(40) Clear-Sky Solar Irradiance	taub		0.336	0.355	0.390	0.415	0.424	0.420	0.424	0.415	0.393	0.371	0.348	0.330		
		taud	2.369	2.352	2.259	2.225	2.250	2.279	2.277	2.319	2.370	2.409	2.381	2.345		
	Ebn at Noon		668	760	797	821	831	837	826	813	789	732	645	605		
		Edh at Noon	64	83	109	127	129	127	125	114	98	78	62	56		
(44) All-Sky Solar Radiation	RadAvg		0.75	1.45	2.69	4.42	5.18	5.59	5.28	4.51	3.30	1.88	0.88	0.57		
	RadStd		0.06	0.26	0.34	0.39	0.37	0.43	0.54	0.37	0.22	0.16	0.09	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-100	N/A	N/A
(47)	Regional (81 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page